

Ex. No: 2 Illustrate UNIX Commands and Shell Programming

1: Greatest Among Three Numbers

Program:

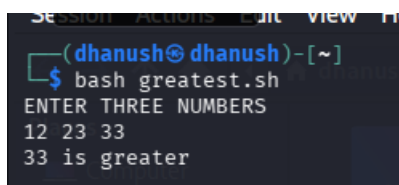
```
#!/bin/bash

echo "ENTER THREE NUMBERS"

read a b c

if [ $a -gt $b ] && [ $a -gt $c ]
then
echo "$a is greater"
elif [ $b -gt $c ]
then
echo "$b is greater"
else
echo "$c is greater"
fi
```

Output:

A terminal window with a dark background. The prompt is (dhanush@dhanush)~. The user enters 'bash greatest.sh'. The script outputs 'ENTER THREE NUMBERS'. The user enters '12 23 33'. The script outputs '33 is greater'.

```
SESSION ACTIONS Edit View File
(dhanush@dhanush)~[~]
$ bash greatest.sh
ENTER THREE NUMBERS
12 23 33
33 is greater
```

2. Factorial of a Given Number

Program:

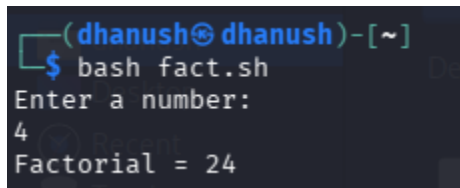
```
#!/bin/bash

echo "ENTER THE NUMBER:"

read n
```

```
fact=1
while [ $n -gt 1 ]
do
fact=$((fact * n))
n=$((n - 1))
done
```

Output:

A terminal window with a dark background. The prompt is '(dhanush@dhanush)~'. The user has run 'bash fact.sh'. The script prompts 'Enter a number:' and the user has entered '4'. The script outputs 'Factorial = 24'.

```
(dhanush@dhanush)~  
$ bash fact.sh  
Enter a number:  
4  
Factorial = 24
```

3. Sum of Odd Numbers up to N

Program:

```
#!/bin/bash
echo "ENTER THE RANGE:"
read n
x=1
sum=0
while [ $x -le $n ]
do
sum=$((sum + x))
x=$((x + 2))
done
```

Output:

```
(dhanush@dhanush)-[~]  
$ bash oddsum.sh  
ENTER THE RANGE:  
22  
SUM = 121
```

4. Generation of Fibonacci Numbers

Program:

```
#!/bin/bash
```

```
echo "ENTER THE LIMIT:"
```

```
read n
```

```
p=-1
```

```
q=1
```

```
i=1
```

```
while [ $i -le $n ]
```

```
do
```

```
r=$((p + q))
```

```
p=$q
```

```
q=$r
```

```
echo "$r"
```

```
i=$((i + 1))
```

```
done
```

Output:

```
(dhanush@dhanush)~[~]  
$ bash fibonacci.sh  
ENTER THE LIMIT:  
4  
0  
1  
1  
2
```

5.Arithmetic Calculator

Program:

```
#!/bin/bash  
  
echo "ENTER THE VALUE OF A:"  
read a  
  
echo "ENTER THE VALUE OF B:"  
read b  
  
echo "ENTER THE OPTION TO PERFORM"  
echo "1. ADDITION"  
echo "2. SUBTRACTION"  
echo "3. MULTIPLICATION"  
echo "4. DIVISION"  
read op  
  
case "$op" in  
1) echo "Result = $((a + b))" ;;  
2) echo "Result = $((a - b))" ;;  
3) echo "Result = $((a * b))" ;;  
4) echo "Result = $((a / b))" ;;  
*)
```

Output:

```
(dhanush@dhanush)-[~]  
$ bash arithmetic.sh  
ENTER THE VALUE OF A:  
3  
ENTER THE VALUE OF B:  
4  
ENTER THE OPTION TO PERFORM  
1. ADDITION  
2. SUBTRACTION  
3. MULTIPLICATION  
4. DIVISION  
1  
Result = 7
```

6.Largest Digit of Number

Program:

```
#!/bin/bash  
  
echo "ENTER THE NUMBER"  
  
read a  
  
max=0  
  
while [ $a -gt 0 ]  
do  
r=$((a % 10))  
if [ $r -gt $max ]  
then  
max=$r  
fi  
a=$((a / 10))  
done  
  
echo "THE LARGEST DIGIT OF THE NUMBER: $max"
```

Output:

```
(dhanush@dhanush)-[~]  
$ bash largestDigit.sh  
ENTER THE NUMBER  
1234  
THE LARGEST DIGIT OF THE NUMBER: 4
```

7. Palindrome String Check

Program:

```
#!/bin/bash  
  
echo "ENTER THE STRING TO CHECK PALINDROME"  
  
read str  
  
len=$(echo -n "$str" | wc -c)  
  
i=1  
  
j=$((len / 2))  
  
while [ $i -le $j ]  
  
do  
  
k=$(echo "$str" | cut -c $i)  
  
l=$(echo "$str" | cut -c $len)  
  
if [ "$k" != "$l" ]  
  
then  
  
echo "$str is not a palindrome"  
  
exit  
  
fi  
  
i=$((i + 1))  
  
len=$((len - 1))  
  
done  
  
echo "$str is a palindrome"
```

Output:

```
(dhanush@dhanush)-[~]  
$ bash palindrome.sh  
ENTER THE STRING TO CHECK PALINDROME  
121  
121 is a palindrome
```

8. Reverse of a Given Number

Program:

```
#!/bin/bash  
  
echo "ENTER THE NUMBER"  
  
read n  
  
rnum=0  
  
while [ $n -ne 0 ]  
do  
    remainder=$((n % 10))  
    rnum=$((rnum * 10 + remainder))  
    n=$((n / 10))  
done  
  
echo "REVERSE OF THE NUMBER IS $rnum"
```

Output:

```
(dhanush@dhanush)-[~]  
$ bash reverse.sh  
ENTER THE NUMBER  
12345  
REVERSE OF THE NUMBER IS 54321
```